INCH-POUND

MIL-I-25324C(USAF)
15 June 02
SUPERSEDING
MIL-I-25324B(USAF)
15 MAY 1974

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, 0-12,000 PPH

This specification is approved for use of all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope this specification covers the requirements for two types of nonintegrally lighted indicators used to indicate the rate at which fuel flows to the engine. The indicators are hermetically sealed, remote second-harmonic selsyn receivers incremented in pounds per hour (pph).

1.2 Classification. The indicators shall be of the following types, as specified (see 6.2):

Type MA-7	Provides an auxiliary summation output signal
Type MA-7A	Same as MA-7 except it does not provide summation output signal.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-1672 Boxes, Shipping, Reusable with Cushioning

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

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Military

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-116	Preservation, Methods of
MIL-I-7566	Indicator System, Fuel Flowmmeters General Specification For
MIL-T-8249	Transmitter, Rate of Flow, Fuel, Type MB-2
MIL-P-8422	Power Supply, 28V DC Input, 15V 4 Cycle 3 Phase Output, Type MC-1
MIL-I-25325	Indicator, Rate of Flow, Fuel, Totalizing, 0-100,000 PPH, Type MA-8

STANDARDS

Federal

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-2073/1A	DOD Material Procedures For Development and Application Of Packaging Requirements
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric, Angle 90°
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption; has been obtained.

3. REQUIREMENT

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2d) in accordance with 4.2.

3.2 General specification. The requirements of MIL-I-7566 apply as

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requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction. The Type MA-7 and Type MA-7A indicators shall be designed and constructed in accordance with figures 1 and 2, respectively.

3.3.1 Individual rate indication. The indicators shall operate on the second harmonic system in which the moving element consists of a nonenergized permanent 2-pole magnet (or magnets) and the torroidally wound stator is energized with 115V 400-cycle single-phase a-c power. In addition to the excitation voltages, even harmonics may appear across the winding, depending upon the position of the moving element.

3.3.2 Summation output signal (Type MA-7). The Type MA-7 indicator shall supply for summation a 400-cycle voltage of 1 volt per 1,000 pph (pounds per hour). The 400-cycle voltage signal may be interrupted at a frequency of 4 cycles per second (cps).

3.3.2.1 Totaling indication. Failure of one or more individual rate indicators shall not affect totalizing of the system, and failure of the totalizing indicator shall not affect normal individual rate of indication of the Type MA-7 indicators.

3.3.3 Calibration. The indicator dials shall be graduated in accordance with figures 1 and 2. The calibration of fuel flow versus angular degrees shall be linear in each range with $\pm$ 0° 15' of arc, except at the changeover point, and from 0 through 400 pph.

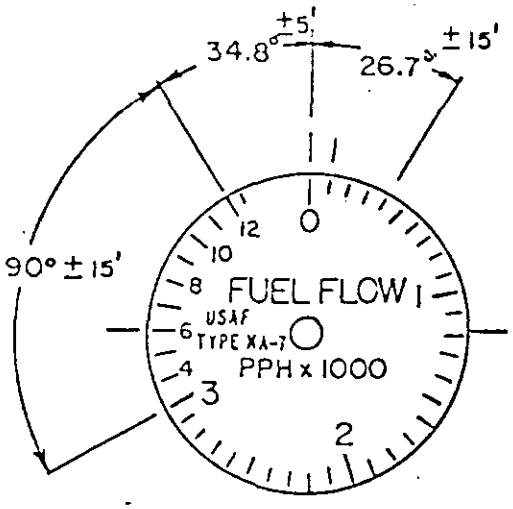
3.3.3.1 Indicator designed. The indicator shall be designed to indicate, at a remote station, the rate of flow (in pounds per hour) of aviation fuel when it is used in conjunction with a fuel flow transmitter conforming to MIL-T-8249, and a power supply conforming to MIL-P-8422. The Type MA-7 indicator provides an output signal to be used with a summation indicator conforming to MIL-I-25325.

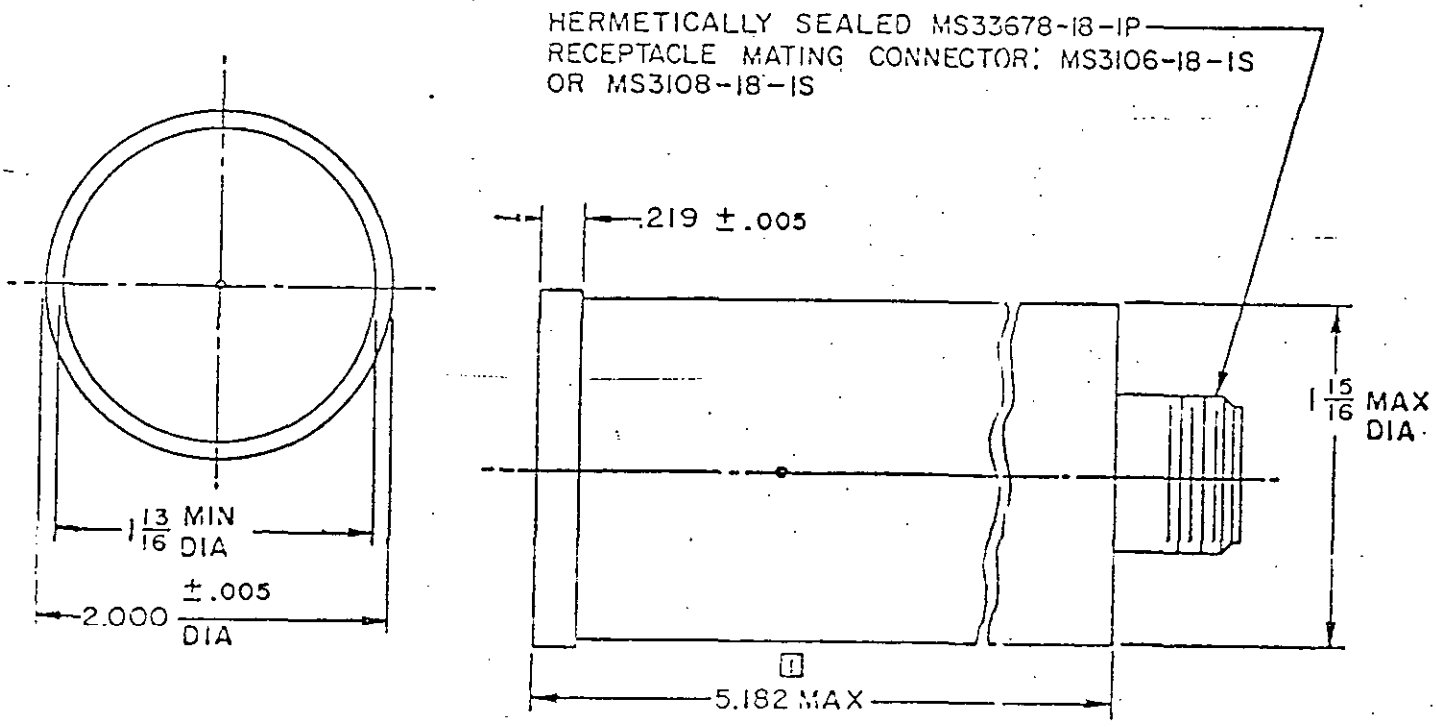
3.4 Pointer. The pointer shall conform to MS33585-8.

3.5 Markings. The following markings shall be finished in yellow fluorescent-luminescent material. The dimensions shall be as specified.

	Height or Length Inch + 1/64	Width of Line or Graduation Inch + 0.005
Scale numerals 0 through 3	3/16	0.031
Scale numerals 4 through 12	1/8	0.020
Numbered 1000-pound graduations 0 through 12	1/8	0.20
Numbered 1000-pound graduations 4 through 12	5/32	0.020
Intermediate 1000-pound graduations	1/8	0.020

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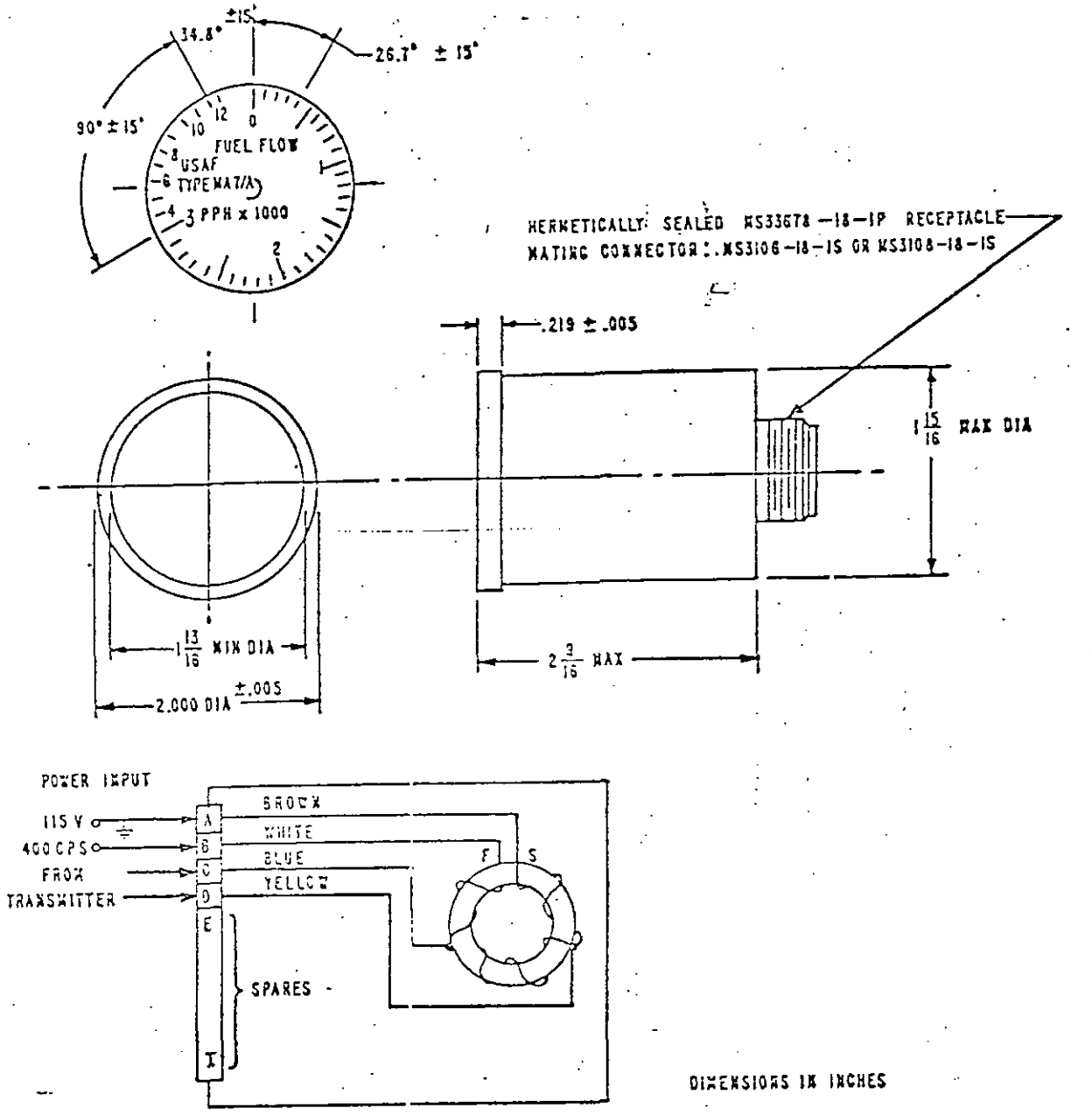
DIMENSIONS IN INCHES

NOTES

1. Background of Dial Dull Black Paint
2. Dial Graduations Linear above 400 PPH with a Scale Change at 3000 PPH

FIGURE 1. MA-7 Indicator Outline

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NOTES

1. Background of Dial: Dull Black
2. Dial Graduations Linear above 400 PPH Mark with a Scale Change at 3000 PPH Mark

FIGURE 2. MA-7A Indicator Outline and Schematic

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500-pound graduations		
0 through 3	1/8	0.031
100-pound graduations	3/32	0.020
Lettering PPH X 1000	3/32	0.016
Lettering Fuel Flow	1/8	0.020
Shaded portion of pointer	---	-----

3.5.1 Legibly marking. Markings USAF and TYPE MA-7 or TYPE MA-7A, as applicable, shall be permanently and legibly marked on the dial in 3/64-inch high letters. These and all other markings for which the color is not otherwise specified shall be finished in durable dull black.

3.5.2 Reference mark. A 500-pph linear graduation shall be placed on the dial at the 12,500-pph position to be used as a reference mark in indicating flow rates in excess of 12,000 pph.

3.6 Dial diameter. The dial shall have a minimum diameter of 1-3/4 inches as measured at the outside ends of the outer scale graduations.

3.6.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be held to a minimum and shall not exceed 0.150 inch.

3.7 Electrical receptacle. The electrical receptacle for the Type MA-7 and Type MA-7A indicators shall be in accordance with figures 1 and 2 respectively.

3.8 Schematic. The wiring schematic for the Type MA-7A and Type MA-7 indicators shall be in accordance with figures 2 and 3, respectively.

3.9 Case. The case shall be hermetically sealed and shall conform to MS33639 for the 2-inch size. Dimension L shall be 5 ± 1/4 inches for the Type MA-7 indicator and shall be 2-9/16 inches for thee Type MA-7A indicators.

3.10 Weight. The weight of the Type MA-7 indicator shall not exceed 1.25 pounds and the weight of the Type MA-7A indicator shall not exceed 0.6 pound.

3.11 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of tests. The inspection and testing of indicators shall be in accordance with MIL-I-7566 and as specified herein. Except where qualification is called out in MIL-I-7566, it shall be understood to be first article inspection.

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4.3 Performance. Performance of the indicators shall be checked at the test points specified in tables I and II and shall not exceed the tolerances specified therein.

4.4 Inspection and test. Test of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to ensure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method 1C-1 IAW MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073/1A to provide a method 1C-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073/1A.

5.2 Packing. Packing shall be level "A", "B", or "C" as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material between the item and unit container to protect the unit container from punctures. Intermediate container requirements shall conform to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, special packaging instruction (SPI), and contracting activity.

5.2.3 Level B. Same as level "A" unless otherwise specified by special

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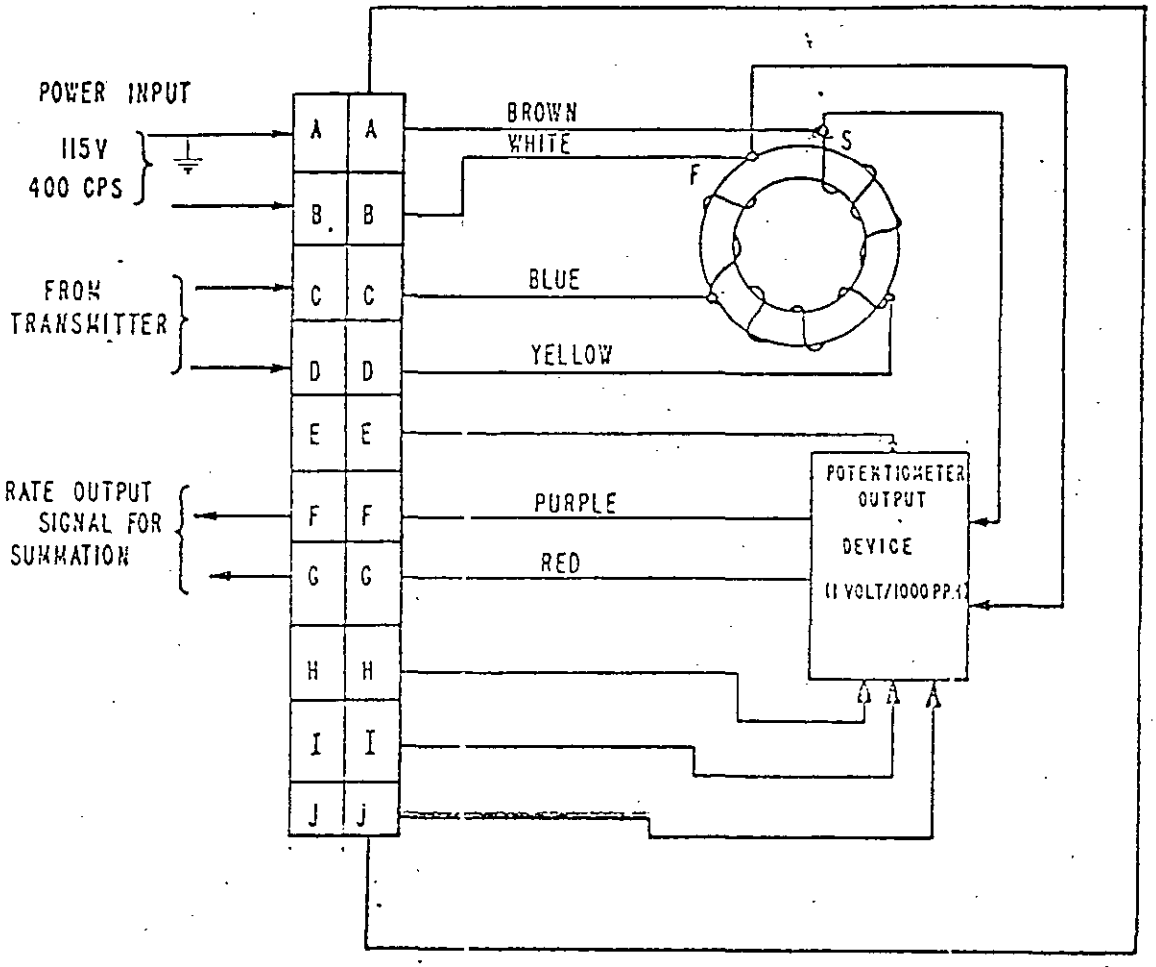


FIGURE 3. MA-7 Indicator Schematic .

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TABLE I

Transmitter Setting (Degrees)	Indicator Reading (pph)	Maximum Indicator Error ±(pph)	Maximum Indicator Error ±(Degrees)	Potentiometer Output Signal (Volts) (MA-7 only)	Max Potentiometer Output Error ±(pph)	Max Potentiometer Output Error ±(Volts)
0	0	0	0	0	45	0.045
11.4	200	10	0.7	0.2	10 to -90	0.010 to 0.090
35.2	500	10	0.8	0.5	45	0.045
75.2	1,000	10	0.8	1.0	45	0.045
115.2	1,500	10	0.8	1.5	45	0.045
155.2	2,000	10	0.8	2.0	45	0.045
195.2	2,500	10	0.8	2.5	45	0.045
235.2	3,000	10	0.8	3.0	45	0.045
245.2	4,000	100	1.0	4.0	360	0.360
265.2	6,000	100	1.0	6.0	360	0.360
285.2	8,000	100	1.0	8.0	360	0.360
305.2	10,000	100	1.0	10.0	360	0.360
325.2	12,000	100	1.0	12.0	360	0.360

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packaging instruction (SPI) or the contracting activity.

5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073/1A.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, special packaging instructions (SPI), and MIL-STD-129.

5.4 Inspection and Test. Test methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116 to ensure compliance with Section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

TABLE II

TEST	TOLERANCE PPH			
	Scale Error		Summation Output Error	
	0-3000 Range	3000-12000 Range	0-3000 Range	3000-12000 Range
Frequency and Voltage Variation	10	100	10	100
position error at 2500 pph	10	-	10	-
Friction				
room and high temperature	10	100	10	100
Low temperature	20	200	20	200
Vibration				
Pointer oscillation	20	200	20	200
Pointer variation	20	200	20	200
Endurance Response time 3 seconds max Power input 6.5 watts max	70	550	15	150

6.1 Intended use. The Type MA-7 indicator covered by this specification is intended for use in indicating the rate of, and to initiate a signal for the

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summation of, the flow of fuel in pounds per hour consumed by aircraft turbojet engine. The TYPE MA-7A indicators is functionally similar to the Type MA-7 indicator except that it does not supply an output signal.

6.2 Ordering data. Procurement document should specify the following:

a. Title, number, and date of this specification.
b. Type of indicator required (see 1.2).
c. Selection of applicable levels of packaging and packing is made by using Section 5.
d. When a first article is required for inspection and approval (see 3.1 and 4.2).
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2.

6.4 Subject term (key word) listing.

Nonintegrally lighted indicator
Rate of flow, fuel

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing Activity:
AIR FORCE -71

Project No:
6620-F489

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-25324C(USAF)	15 May 1974

3. DOCUMENT TITLE

Indicator, Rate Of Flow, Fuel, 0-12,000 PPH

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

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